find me a youtube tuortial and then give me a test tutorial on blender animating a skeleton and exporting to godot.

i want a short video that explains everything as if Im a beginner. assume I already have a 3d model

Here's a good **beginner-friendly YouTube tutorial** to start with, followed by a **step-by-step text tutorial** you can follow after watching it:

This short video walks through *basic rigging*, creating weights, a simple animation, and exporting into **Godot 4 —** great if you want everything shown *visually* and at a beginner pace. YouTube

---

🦴 Quick Text Tutorial: Blender → Godot (Assuming You Have a 3D Model)

1. Set up an Armature (Skeleton) in Blender

1. **Open your model in Blender.**

2. Go to the **Add** → **Armature** → **Single Bone** to create the root bone.
3. Enter **Edit Mode** (Tab) and **extrude bones** (select bone, press **E**) to match parts of your mesh (spine, arms, legs).
4. Make sure the bones form a proper hierarchy (parent → child chain).
   (This aligns bones so they deform your mesh correctly.)

Tip: In Edit Mode, enable *In Front* display so you can see the bones through the mesh.

---

2. Bind Skeleton to the Mesh

1. Select your mesh, then select the armature (Shift+click).
2. Press **Ctrl + P** → **With Automatic Weights** to bind the model to the bones.
   This assigns how each bone influences parts of the mesh.

Tip: You can refine joint influence in **Weight Paint Mode** if some areas deform weirdly.

Games in Progre...

---

3. Animate Your Rig

1. Switch to **Pose Mode** (select armature).
2. Open the **Timeline / Dope Sheet**.
3. Move to frame 1 → pose bones → insert keyframes (**I** → **Loc/Rot/Scale**).
4. Go to later frames → adjust bones → insert keyframes again.
5. Make *multiple actions* (walk, idle, etc.) by creating and naming animations separately.

Use the **NLA Editor** to organize multiple clips. Games in Progre...

---

4. Export for Godot

1. **Select your mesh + armature.**
2. **File → Export → glTF (.glb/.gltf)**
   - *Choose* `.glb` for a single self-contained file.
   - Make sure **Include** → **Selected Objects** is checked.
3. In export settings ensure **Animation** and **Skinning** options are enabled (default for glTF).
   Godot prefers glTF for skeletal animations.

5. Import into Godot

1. Copy the exported **.glb** into your Godot project.
2. Godot will auto-import the model with animations.
3. In the **Scene Tree**, you'll see your model with a **Skeleton3D** + **AnimationPlayer / AnimationTree**.
4. **Test play** by selecting animations in the AnimationPlayer or set up an **AnimationTree** to blend walk/idle etc.

🧠 Extra Tips

- **Use glTF (.glb)** — best compatibility with Godot.
- **Avoid complex control bones** for export; Godot may not interpret them — stick to forward kinematics or bake animations.
- Match Godot's expectations by keeping bone names consistent.
- You *can reuse animations across characters* in Godot with BoneMap retargeting (Godot 4 feature). Godot Engine d...

If you want, I can also provide **a sample Blender file setup** you can download or a **Godot scene example** that shows a working animated character ready to test in Godot.

Sources

I have a 3d platformer working in a 2d plane. So it plays like a 2d game, but i'm using 3d models.

I have a chracter that carries a bow. I want them to be able to shoot and run at the same time. When they shoot they should be aiming at the player cursor.

how do I do this.
is it a mix of inverse kinematics and animation? or is it blending two animations together?

i'm not sure of the workflow to make this happen

Great — this is a classic **2.5D character animation problem** (3D model, 2D gameplay plane).
And yes — what you want is a **combination of animation blending + IK**.

Not just one or the other.

Here's the correct mental model and workflow 👇

🎯 What You're Trying To Achieve

You want:

- ✅ Character can **run**
- ✅ Character can **aim at cursor**
- ✅ Character can **shoot while running**
- ✅ Upper body rotates independently from lower body
- ✅ Bow/arms track mouse position dynamically

This is done with:

- **Base locomotion animation (run/idle)**
- **Upper body aim animation layer**
- **IK to precisely align hands to bow / arrow**

🧠 The Correct Setup (Industry Standard Way)

You need 3 systems working together:

1️⃣ Lower Body = Movement Animation

- Idle
- Run
- Jump

These animations affect:

- Hips
- Legs
- Root motion (optional)

2️⃣ Upper Body = Aim Pose (Animation Blend Space)

Instead of animating every possible aim angle manually...

You create:

- An **Aim Pose** (arms holding bow forward)
- Then use:
  - **Bone rotation from script**
  - OR
  - **Aim blendspace**
  - OR
  - **IK target tracking cursor**

3️⃣ IK = Make Hands Stick to Bow Correctly

IK is used for:

- Making left hand stay on bow
- Making right hand pull string
- Making arms follow cursor smoothly

🎮 Since You're Using Godot

In **Godot 4**, you use:

- `AnimationTree`
- `AnimationNodeBlendTree`

- `AnimationNodeBlendSpace1D or 2D`
- `SkeletonIK3D`

🔥 Recommended Workflow (Beginner-Friendly)

STEP 1 — Animate in Blender

Create these animations:

1. **Idle**
2. **Run**
3. **Aim Pose (static pose)**
    - Arms holding bow straight forward
    - No leg movement

Export as `.glb`.

STEP 2 — Set Up AnimationTree in Godot

In your character scene:

Add:

- `AnimationPlayer`
- `AnimationTree`

Set AnimationTree → Active = ON

Set Tree Root → `AnimationNodeBlendTree`

STEP 3 — Blend Lower Body and Upper Body

Use:

➤ `AnimationNodeBlend2`

Base animation = Idle/Run

Blend animation = Aim Pose

But here's the trick:

Use **Bone Filters** so only:

- Spine
- Chest
- Arms
- Head

are affected by the Aim animation.

This allows:

Lower body → keeps running
Upper body → aims independently

This is called **layered animation blending**.

🎯 STEP 4 — Make Character Aim at Mouse

Now we add logic.

Because your game is 2.5D:

Movement plane is likely:

- X (left/right)
- Y (up/down)
- Z locked

So aiming becomes:

```
var world_pos = camera.project_ray_origin(mouse_pos)
```

gdscript

Then:

Rotate upper spine bone toward direction.

Option A (Simpler) — Rotate Spine Bone Directly

Get spine bone index:

gdscript

Apply rotation override:

```
    bone_idx,
    new_transform,
    1.0,
```

gdscript

Option B (Better) — Use SkeletonIK3D

Add:

Code

Set:

- Root bone = shoulder
- Tip bone = hand

Create a target `Node3D`

Move that target to cursor world position.

Now arm follows cursor automatically.

🏹 STEP 5 — Shooting While Running

Shooting animation should be:

- Only upper body
- Short animation (draw → release)

You blend it on top of run animation using:

```
    ↓
UpperBodyAim
```

Code

In AnimationTree:

Use:

- `AnimationNodeOneShot` for shooting
- Trigger it from script

gdscript

🧩 So Is It IK or Blending?

Correct answer:

Feature	Solution
Running + shooting	Animation blending
Aiming at cursor	Bone rotation or IK
Keeping hands aligned to bow	IK
Playing shoot animation	OneShot animation node

It’s a **layered system**.

🏗️ Clean Mental Model

Think in layers:

```
Upper Layer → Aim Direction
```

Code

This is how AAA games do it.

⚠️ Common Beginner Mistake

Trying to:

- Make a separate animation for every angle
- Or animate run+shoot as one animation

Don’t do that.

You want modular animation layers.

🧠 If You Want Simplest Possible Version

If you're early in development:

1. Play run animation normally
2. Rotate entire upper body bone toward mouse
3. Spawn arrow toward mouse
4. Skip IK at first

Add IK polish later.

🏁 Final Answer

Yes — it’s a mix.

✅ Animation blending handles movement + shooting
✅ IK (or bone rotation) handles aiming
✅ AnimationTree controls the layers
✅ Script feeds mouse direction into skeleton

If you'd like, I can:

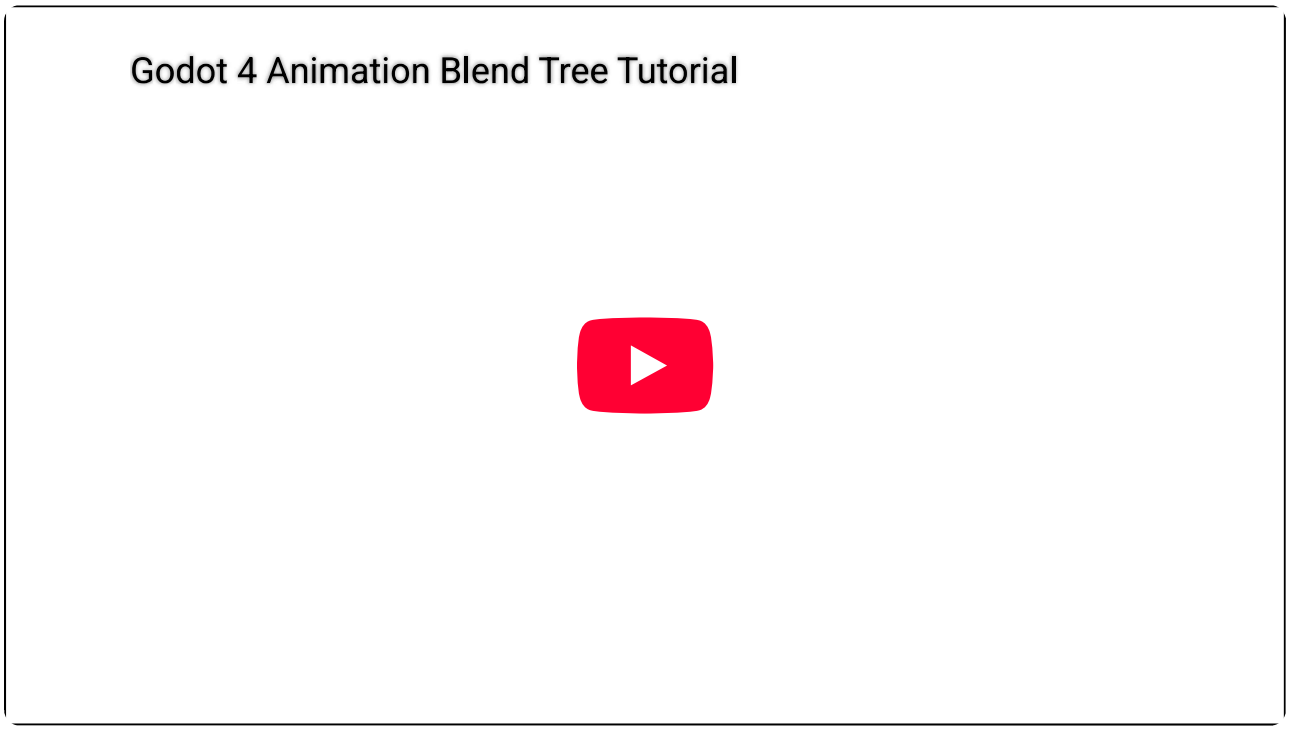
- Draw you a node diagram for your AnimationTree
- Or write a minimal working Godot 4 script setup for this system

Which would help more right now?

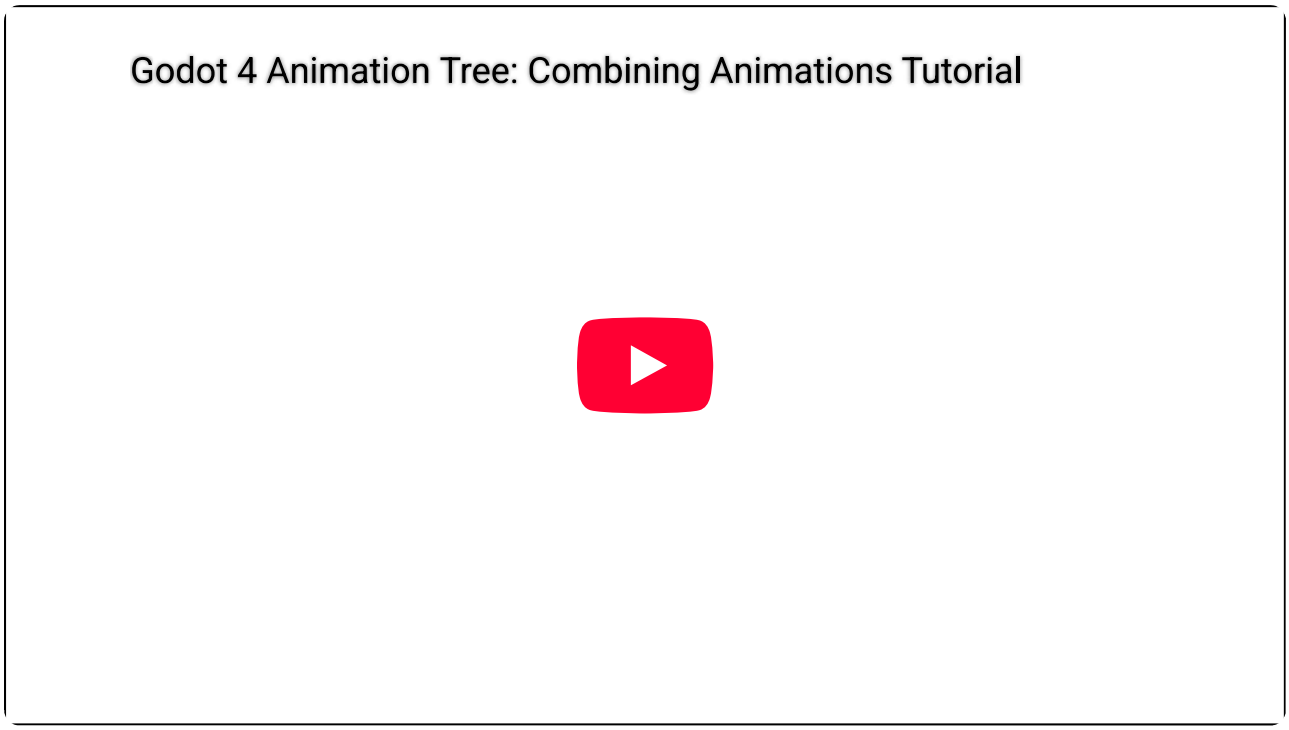
Can you link me a youtube video that explains thbis

Here are **YouTube videos you can start with** that will help you understand the *core systems* you need for blending and aiming:

📌 Animation Blending in Godot 4 (Beginner-Friendly)



This video walks through how to use Godot 4's **AnimationTree blend tree** system to mix multiple animations (like idle & run). It's *perfect* for learning how to layer upper-body animations over movement. YouTube



Another clear step-by-step showing how to combine animations using AnimationTree — useful for rigging up *walk* + *aim* blending. YouTube

---

🎯 IK (Inverse Kinematics) Basics in Godot 4

This one goes over the **IK system in Godot 4**, which you'll need to have the character's arms/hand actually *track aim targets* dynamically. YouTube

---

🧠 How to Use These Together

Once you've watched those two types of videos:

✅ **AnimationTree** → lets you blend your lower-body run/idle animations with an upper-body aim animation.
✅ **IK** → lets you make the arms/bow *orient toward the cursor* during gameplay, beyond what static animations can do.

ChatGPT Log in Sign up for free

1. **Import your rig and animations**
2. **Set up an AnimationTree** with blend nodes for run/aim layers
3. **Enable bone filters** so only upper body is affected by aim layer
4. **Add IK on the arm chain** so the hand tracks the cursor target
5. **Write GDScript** to move the IK target to the mouse world position

---

If you want **a video specifically about aiming weapons while running (upper body blend + IK)** I can find a more focused one — just tell me whether you prefer **Godot 3 or Godot 4** tutorials. (Most new ones are for Godot 4.)

Sources

give me a prompt to recreate this chat

Here's a clean prompt you can reuse to recreate this exact type of conversation:

Prompt:

> I'm making a 2.5D game in Godot (3D models but gameplay constrained to a 2D plane).
>
> I have a character that carries a bow. I want the character to be able to:
>
> - **"Run and shoot at the same time"**
> - **"Aim toward the mouse cursor"**
> - **"Blend running animation with upper-body aiming"**
> - **"Possibly use IK for arm positioning"**
>
> Assume I already have a rigged 3D model with animations made in Blender.
>
> Please explain:
>
> 1. **"The correct animation workflow (Blender → Godot)"**
> 2. **"Whether this requires animation blending, IK, or both"**
> 3. **"How to structure the AnimationTree"**
> 4. **"How to aim at the cursor in a 2.5D setup"**
> 5. **"Common beginner mistakes to avoid"**
>
> Explain it clearly like I'm still learning but want to understand the proper game dev workflow.

If you'd like, I can also give you:

- A **shorter version** of the prompt

- A **more technical version**
- Or a version tailored specifically for Blender, Unity, or Unreal instead of Godot